

Task 18: Deploy Twenty CRM on Kubernetes

Objective

The objective of this task was to deploy the Twenty CRM application on a local Kubernetes cluster using Docker Desktop Kubernetes. A Kubernetes Deployment and Service were created, the application was verified through the Service, the Deployment was scaled to two replicas and back to one replica, and the Kubernetes resources were removed after testing.

Environment

- Operating System: Windows
- Kubernetes platform: Docker Desktop Kubernetes
- Kubernetes context: docker-desktop
- Kubernetes version: v1.36.1
- kubectl client version: v1.36.1
- Docker Engine: 29.7.2
- Git branch: bhavish-18

Kubernetes Setup

Docker Desktop Kubernetes was enabled and used as the local Kubernetes environment. The cluster was verified before deploying Twenty CRM.

```
kubectl get nodes
```

The Kubernetes node was reported in the Ready state.

```
kubectl cluster-info
```

The Kubernetes control plane and cluster services were successfully available.

Project Structure

The Kubernetes configuration was added to the repository using the following structure:

```
devops-crm-project/  
├── TASK-18.md  
├── TASK-18-DOCUMENTATION.docx  
└── kubernetes/  
    └── twenty-crm.yaml
```

Kubernetes Configuration

A Kubernetes manifest named `twenty-crm.yaml` was created. The configuration contains a Deployment for the Twenty CRM application and a NodePort Service for application access.

Twenty CRM Docker Image

The existing Twenty CRM development image was used:

```
twentycrm/twenty-app-dev:latest
```

Deployment

The Deployment is named `twenty-crm`. It initially runs one replica and uses the Twenty CRM container image. The container exposes port 2020.

```
replicas: 1
containerPort: 2020
```

Environment Variables

The Kubernetes Deployment uses the development environment configuration from the provided Twenty CRM reference:

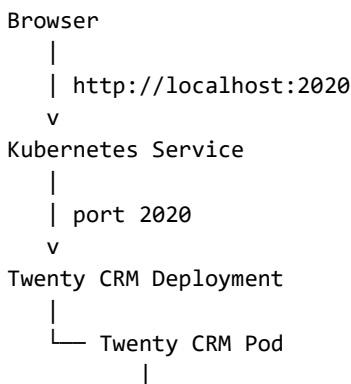
- NODE_PORT: 2020
- SERVER_URL: http://localhost:2020
- STORAGE_TYPE: local
- APP_SECRET: twenty-app-dev-secret-not-for-production
- NODE_ENV: development
- DISABLE_DB_MIGRATIONS: false
- DISABLE_CRON_JOBS_REGISTRATION: false
- IS BILLING_ENABLED: false
- SIGN_IN_PREFILLED: true
- APPLICATION_LOG_DRIVER: CONSOLE

Service

A Kubernetes Service named `twenty-crm-service` was created with type `NodePort`. It exposes port 2020 and forwards traffic to the Twenty CRM container on port 2020.

```
type: NodePort
port: 2020
targetPort: 2020
```

Architecture



└─ Twenty CRM application
Container Port: 2020

YAML Validation

Before deploying the resources, the Kubernetes manifest was validated using a client-side dry run:

```
kubectl apply --dry-run=client -f twenty-crm.yaml
```

The command returned:

```
deployment.apps/twenty-crm created (dry run)
service/twenty-crm-service created (dry run)
```

This confirmed that the Kubernetes manifest was accepted by the client-side validation.

Deployment

The Kubernetes resources were created using:

```
kubectl apply -f twenty-crm.yaml
```

The command returned:

```
deployment.apps/twenty-crm created
service/twenty-crm-service created
```

Verification

The Deployment, Pod, and Service were checked using:

```
kubectl get deployments
kubectl get pods
kubectl get services
```

The Twenty CRM pod initially entered the ContainerCreating state and then successfully changed to Running.

```
twenty-crm-5d55d5475f-9ghlj 1/1 Running 0
```

The pod description confirmed that the container was Running and Ready, with zero restarts. The Kubernetes events showed the successful sequence of scheduling, image pulling, container creation, and container startup.

Service Access

The Service was verified using:

```
kubectl get service twenty-crm-service
```

The Service was exposed through NodePort 30545 and mapped to application port 2020.

```
twenty-crm-service NodePort ... 2020:30545/TCP
```

For local access, port forwarding was used:

```
kubectl port-forward service/twenty-crm-service 2020:2020
```

The application was accessed in the browser at:

```
http://localhost:2020
```

The Twenty CRM interface loaded successfully in the browser. This confirmed that the application was accessible through the Kubernetes Service.

Scaling to 2 Replicas

The Deployment was scaled from one replica to two replicas using:

```
kubectl scale deployment twenty-crm --replicas=2
```

The Deployment was then verified using:

```
kubectl get deployments
```

The result showed:

```
twenty-crm 2/2 2 2
```

The pods were verified using:

```
kubectl get pods
```

Both Twenty CRM pods were reported as 1/1 Ready and Running. This completed the required two-replica scaling test.

Scaling Back to 1 Replica

The Deployment was scaled back to one replica using:

```
kubectl scale deployment twenty-crm --replicas=1
```

Verification was performed using:

```
kubectl get deployments  
kubectl get pods
```

The final Deployment state showed one available replica and one Running pod.

```
twenty-crm 1/1 1 1
```

Cleanup

After completing deployment, application access, and scaling verification, the Kubernetes resources were deleted as required by the task.

```
kubectl delete -f kubernetes/twenty-crm.yaml
```

The command returned:

```
deployment.apps "twenty-crm" deleted from default namespace
service "twenty-crm-service" deleted from default namespace
```

The resources were then verified using:

```
kubectl get deployments
kubectl get pods
kubectl get services
```

The Deployment and Pods returned:

```
No resources found in default namespace.
```

The default Kubernetes Service remained in the cluster. This is the standard Kubernetes service and was left unchanged.

Git Deliverables

The task was performed on the required branch:

```
bhavish-18
```

The files prepared for the task are:

- TASK-18.md
- TASK-18-DOCUMENTATION.docx
- kubernetes/twenty-crm.yaml

The files can be added and committed using:

```
git add TASK-18.md TASK-18-DOCUMENTATION.docx kubernetes/twenty-crm.yaml

git commit -m "Add Kubernetes deployment for Twenty CRM"

git push -u origin bhavish-18
```

After pushing the branch, a Pull Request should be raised in the devops-crm-project repository.

Result

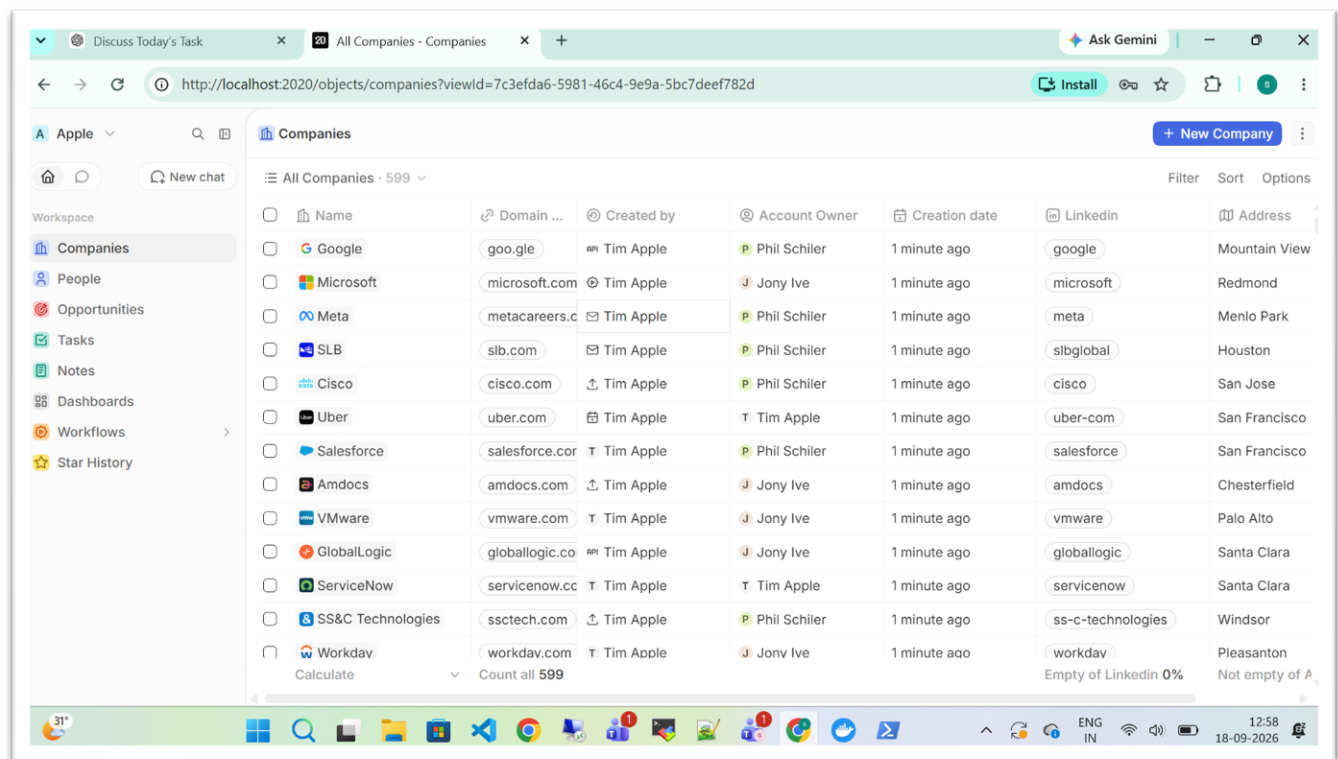
- Docker Desktop Kubernetes cluster was successfully configured and verified.
- Twenty CRM was deployed using a Kubernetes Deployment.
- A NodePort Service was created for application access.
- The Twenty CRM pod reached the Running and Ready state.
- Twenty CRM was successfully accessed at <http://localhost:2020>.
- The Deployment was successfully scaled to 2 replicas.
- Both replicas were verified as Running.
- The Deployment was scaled back to 1 replica.
- The Kubernetes Deployment and Service were deleted after testing.

Conclusion

Task 18 successfully demonstrated deployment of the Twenty CRM application on a local Kubernetes cluster using Docker Desktop Kubernetes. The Kubernetes Deployment and Service were created and verified, the application was accessed locally, and Deployment scaling was demonstrated from one replica to two replicas and back to one replica. The Kubernetes resources were deleted after completing the required verification.

Screenshots

Add the screenshots from the completed task below. Suggested screenshots include the Kubernetes node status, YAML validation, successful deployment, running pod, Service, Twenty CRM browser page, two-replica scaling, one-replica verification, and cleanup.



```
Administrator: Windows PowerShell

ode                LastWriteTime             Length Name
-----
a-----          18-09-2026      12:52           1405 twenty-crm.yaml

S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl apply --dry-run=client -f twenty-crm.yaml
deployment.apps/twenty-crm created (dry run)
service/twenty-crm-service created (dry run)
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl apply -f twenty-crm.yaml
deployment.apps/twenty-crm created
service/twenty-crm-service created
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
twenty-crm          0/1     1             0            14s
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl get pods
NAME                READY   STATUS              RESTARTS   AGE
twenty-crm-5d55d5475f-9ghlj  0/1     ContainerCreating    0            14s
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl get services
NAME                TYPE        CLUSTER-IP   EXTERNAL-IP   PORT(S)          AGE
kubernetes          ClusterIP   10.96.0.1     <none>         443/TCP           41m
twenty-crm-service  NodePort    10.96.54.17   <none>         2020:30545/TCP   15s
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl get pods -w
NAME                READY   STATUS              RESTARTS   AGE
twenty-crm-5d55d5475f-9ghlj  0/1     ContainerCreating    0            53s
twenty-crm-5d55d5475f-9ghlj  1/1     Running              0            59s
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl get pods
NAME                READY   STATUS              RESTARTS   AGE
twenty-crm-5d55d5475f-9ghlj  1/1     Running              0            81s
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl describe pod twenty-crm-5d55d5475f-9ghlj
Name:                twenty-crm-5d55d5475f-9ghlj
Namespace:            default
Priority:              0
Service Account:      default
Node:                 desktop-control-plane/172.18.0.2
Start Time:           Fri, 18 Sep 2026 12:53:55 +0530
Labels:               app=twenty-crm
                     pod-template-hash=5d55d5475f
Annotations:          <none>
Status:               Running
IP:                   10.244.0.5
```

```
Administrator: Windows PowerShell

S C:\msys64\home\ADMIN\devops-internship\devops-crm-project\kubernetes> kubectl port-forward service/twenty-crm-service 2020:2020
Forwarding from 127.0.0.1:2020 -> 2020
Forwarding from [::1]:2020 -> 2020
Handling connection for 2020
Handling connection for 2020
Handling connection for 2020
Handling connection for 2020
Handling connection for 2020
Handling connection for 2020
Handling connection for 2020
9918 12:58:30.534265    2636 portforward.go:489] "Unhandled Error" err="error copying from remote stream to local connection: readfrom tcp6 [::1]:2020->[::1]:56128: write
tcp6 [::1]:2020->[::1]:56128: wsasend: An established connection was aborted by the software in your host machine."
```

```
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
twenty-crm-5d55d5475f-9ghlj        1/1     Running   0           10m
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> kubectl scale deployment twenty-crm --replicas=2
deployment.apps/twenty-crm scaled
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
twenty-crm  2/2      2             2           69m
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
twenty-crm-5d55d5475f-6wwq7        1/1     Running   0           11s
twenty-crm-5d55d5475f-9ghlj        1/1     Running   0           69m
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> kubectl scale deployment twenty-crm --replicas=1
deployment.apps/twenty-crm scaled
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> kubectl get deployments
NAME    READY   UP-TO-DATE   AVAILABLE   AGE
twenty-crm  1/1      1             1           74m
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
twenty-crm-5d55d5475f-9ghlj        1/1     Running   0           74m
S C:\msys64\home\ADMIN\devops-internship\devops-crm-project> _
```

32°

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